

Veridian: Hardware-Attested Fractional Financing of Decentralized Solar Infrastructure for Grid-Parity Bypass in South Africa

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Abstract—South Africa’s electricity distribution system is characterised by chronic load-shedding, capital-intensive grid infrastructure, and a structural inability to extend reliable service to the estimated 14.7 million households that lack dependable access. Distributed solar photovoltaic (PV) generation offers a technically viable alternative; however, retail financing constraints and the absence of institutional compliance mechanisms prevent mass adoption.

This paper presents *Veridian*, a decentralized physical infrastructure network (DePIN) protocol that tokenizes cash-flowing solar installations in South Africa under an Equipment-as-a-Service (EaaS) model. The protocol combines three mechanisms: (i) hardware-attested telemetry from TPM 2.0 solar meters that produce cryptographically signed generation records, (ii) ERC-3643 compliant fractional security tokens whose transfers are gated by on-chain identity verification, and (iii) deterministic USDC yield settlement executed on Optimism Sepolia via a 3-of-5 multisignature escrow. A strict revenue waterfall allocates proceeds as follows: 75% to token holders, 8% to operations and maintenance, 7% to an insurance reserve, 5% to a pool expansion fund, and 5% to platform operations. An end-to-end simulation on a representative 450 W panel at 5.5 sun-hours per day produces a verifiable net yield of 20.7% per annum, which the smart contract suite distributes automatically to verified fractional owners without custodial intermediation. The architecture is evaluated for its ability to disintermediate Eskom’s centralized grid monopoly, reduce household energy costs by at least 10% relative to regulated tariffs, and provide retail investors with compliant, yield-bearing exposure to physical solar infrastructure.

Index Terms—blockchain, decentralized physical infrastructure, DePIN, solar energy, South Africa, equipment-as-a-service, ERC-3643, real-world asset tokenization, TPM 2.0, hardware attestation, yield distribution, Optimism, USDC, multisig

I. INTRODUCTION

South Africa’s electricity sector presents a confluence of structural failure and latent opportunity that is well-suited to blockchain-enabled infrastructure finance. Eskom Holdings SOC Ltd., the vertically integrated state-owned utility, controls the overwhelming majority of generation, transmission, and distribution capacity. Since 2007, Eskom has implemented rolling load-shedding protocols, reaching Stage 6 curtailment in 2022 and 2023, with cumulative lost economic output

estimated by the South African Reserve Bank at over R900 billion [1]. The World Bank estimates that 14.7 million South Africans lack reliable electricity access, with peri-urban and rural communities disproportionately affected [2].

The solar resource across South Africa averages between 4.5 and 6.5 kWh m⁻² day⁻¹ of Global Horizontal Irradiance (GHI), placing the country among the highest-irradiance jurisdictions globally [3], [4]. This resource is geographically distributed and does not require transmission infrastructure to reach end-users when harnessed at the household or community scale. Rooftop and ground-mounted solar PV systems of 450 W to 10 kW capacity are technically adequate to offset the average South African household consumption of approximately 897 kWh per month [5].

Despite favorable resource conditions, retail solar adoption is constrained by two structural barriers. First, the upfront capital cost of a complete 450 W panel system (approximately R6,750 in 2024) is prohibitive for the median South African household income. Second, retail investors seeking exposure to distributed solar infrastructure have no regulated, liquid, and transparent mechanism to participate without deploying project-scale capital.

The Equipment-as-a-Service (EaaS) model addresses the first barrier by converting the capital expenditure of solar acquisition into a per-unit consumption fee paid by the household, analogous to a utility subscription. Prior work on peer-to-peer energy trading [6], [7] has demonstrated the technical feasibility of blockchain-mediated energy markets; however, these systems typically address bilateral settlement between grid-connected prosumers rather than providing a compliant fractional financing layer for the physical equipment itself.

Decentralized Physical Infrastructure Network (DePIN) protocols represent a recent architectural pattern that coordinates the deployment and operation of physical infrastructure through on-chain incentive structures [8]. Applied to energy infrastructure, a DePIN protocol can theoretically align the economic interests of equipment investors, field operators, and energy consumers without a central coordinating entity. Two specific research gaps remain open in this domain. First,

existing DePIN energy protocols (e.g., Power Ledger, Solar-Coin) do not incorporate hardware-level attestation of physical generation data; they rely on software-reported metrics or grid-metered totals, creating an unresolved oracle problem at the physical layer. Second, no published implementation combines compliance-gated fractional securities (as required for regulated investment products) with automated yield settlement in a developing-market context where centralized grid reliability is structurally inadequate.

This paper addresses both gaps through two primary contributions:

- 1) **End-to-end trust chain from physical attestation to regulated settlement.** The Veridian protocol specifies a verifiable pipeline in which TPM 2.0 hardware-signed telemetry propagates through an off-chain yield engine to an ERC-3643 compliant on-chain distribution contract. The architectural contribution is the formal linkage between physical hardware attestation and compliance-gated token settlement, which no existing DePIN protocol provides.
- 2) **Oracle-referenced governance constraint for consumer protection.** The protocol introduces an on-chain price cap mechanism ($P_{\text{proposed}} \leq 0.90 \times P_{\text{grid}}$) that prevents investor-captured DAOs from imposing extractive pricing on energy consumers. This is a novel governance design contribution that addresses an unsolved incentive alignment problem in DePIN protocol governance.

Supporting these contributions, the paper provides: (a) a parameterized economic model of a 450 W South African solar installation producing a base-case net yield of 20.7% per annum with sensitivity analysis across irradiance and degradation scenarios, (b) a formal definition of the 75/8/7/5/5 revenue waterfall and 3-of-5 multisignature escrow mechanics, and (c) a gas cost analysis comparing push and pull distribution patterns on Optimism.

The remainder of this paper is structured as follows. Section II reviews related work on blockchain-based energy systems and DePIN architectures. Section III describes the three-layer Veridian protocol architecture. Section IV presents the economic model and revenue waterfall. Section V describes the implementation and simulation results. Section VI evaluates the system against security, scalability, and compliance requirements. Section VIII concludes.

II. BACKGROUND AND RELATED WORK

A. Blockchain-Based Energy Markets

Andoni et al. [7] provide a systematic review of blockchain applications in the energy sector, identifying peer-to-peer energy trading, grid management, and carbon credit tokenization as the dominant use-cases. Power Ledger, one of the earliest production deployments, enables bilateral energy trading between grid-connected prosumers using a dual-token model. However, Power Ledger and similar systems presuppose functioning grid interconnection and do not address the capital formation problem for equipment acquisition.

Pop et al. [6] evaluate Ethereum-based peer-to-peer energy trading in microgrid environments and identify settlement finality and gas cost as primary constraints for high-frequency micro-transactions. This observation motivates the use of a Layer-2 execution environment in Veridian, as discussed in Section III.

B. Decentralized Physical Infrastructure Networks

Zhang et al. [8] define DePIN protocols as systems that coordinate the deployment of physical hardware through on-chain token incentives, where hardware owners contribute capacity in exchange for protocol-issued rewards. The key design challenge identified in this work is the oracle problem: verifying that physical capacity was actually deployed and utilized without introducing centralized attestation intermediaries. Veridian addresses this challenge through hardware-level attestation rather than software-level oracle aggregation.

C. Real-World Asset Tokenization

Massaro et al. [9] review the regulatory architecture of real-world asset (RWA) tokenization, finding that the principal barriers to adoption are (i) the enforcement of investor eligibility constraints at the token transfer layer and (ii) the linkage between on-chain token state and off-chain legal title. The ERC-3643 standard [10] directly addresses the first barrier through its compliance module and identity registry architecture. The Special Purpose Vehicle (SPV) structure addresses the second.

D. IoT Data Integrity and Hardware Attestation

Boudguiga et al. [11] demonstrate that blockchain anchoring of IoT device firmware updates provides a tamper-evident audit trail that is computationally infeasible to forge without access to the device's private signing key. The Trusted Computing Group's TPM 2.0 specification [12] extends this principle to a standardized hardware security module that can be embedded in field devices to produce cryptographically signed operational attestations. In Veridian, this mechanism is applied to solar generation meters to produce signed kWh telemetry that is verifiable without trusting the metering hardware vendor.

III. PROTOCOL ARCHITECTURE

The Veridian protocol is organized into three layers: the hardware attestation layer, the off-chain oracle and data layer, and the on-chain execution layer. Fig. 1 illustrates the interaction sequence across these layers.

A. Hardware Attestation Layer

Each solar installation is provisioned with a dedicated metering unit containing a TPM 2.0 security module or a software-simulated equivalent operating on the SECP256R1 elliptic curve. During installation, the administrator registers the meter's public key on the off-chain backend through a cryptographic handshake that establishes the Hardware Root of Trust (HROt).

At regular intervals, the meter produces a telemetry payload containing the cumulative energy generated in kWh, a Unix

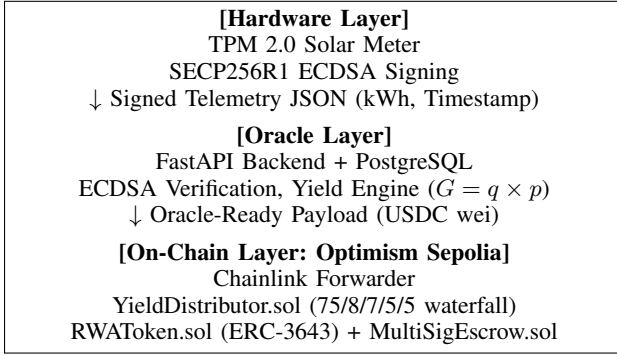


Fig. 1. Veridian three-layer architecture. Physical generation data flows from the signed hardware telemetry through the yield engine to the on-chain distribution contract.

timestamp, and the meter identifier. This payload is signed using the TPM’s internal ECDSA private key, which is hardware-protected and cannot be extracted. The signed JSON object is transmitted to the FastAPI backend:

$$\sigma = \text{ECDSA}_{\text{SECP256R1}}(k_{\text{priv}}, H(\text{kWh}||\text{ts}||\text{id})) \quad (1)$$

where k_{priv} is the TPM-resident private key, $H(\cdot)$ is a SHA-256 hash function, and ts is the Unix timestamp.

The backend verifies σ against the registered public key k_{pub} . Any tampered or replayed payload fails verification and is rejected before database insertion. This mechanism ensures that the yield calculation is grounded in cryptographically verifiable physical generation records rather than operator-supplied data.

A production guard on the variance between hardware-signed utilization records and physical maintenance audit findings is set at a maximum of 5%. If discrepancy exceeds this threshold, the operator’s staked deposit is subject to slashing via the escrow contract.

B. Off-Chain Oracle and Yield Engine

The oracle layer consists of a FastAPI application backed by a PostgreSQL database. The database schema maintains four primary relations: `assets` (token address, SPV status, jurisdiction), `billing_cycles` (computation periods), `telemetry_logs` (verified generation history), and `yield_calculations` (gross and net yield breakdowns).

The yield engine computes gross revenue per billing cycle as:

$$G = q \times p \quad (2)$$

where q is the total energy consumed (kWh) by the lessee household over the billing cycle, and p is the contracted EaaS per-unit rate (ZAR/kWh). The per-unit rate is constrained on-chain by the price cap mechanism described in Section IV-F. The engine then applies the 75/8/7/5/5 waterfall to produce per-tranche allocations and formats the net investor yield as an ABI-encoded payload denominated in USDC-wei (18 decimal precision) for consumption by the `YieldDistributor` contract.

The oracle endpoint `/yields/oracle/{asset_id}` is authenticated via a rotation-based API key linked to the Chainlink Functions subscription. This endpoint returns a deterministic payload that the Chainlink Forwarder submits to the on-chain `onReport()` function.

C. On-Chain Execution Layer (Optimism Sepolia)

Optimism Sepolia was selected as the execution environment for three reasons. First, the Optimistic Rollup architecture inherits the security properties of Ethereum Layer 1 through fraud proofs, while reducing transaction fees by approximately two orders of magnitude. Second, the near-zero cost of micro-transactions on Optimism makes monthly yield distributions to large numbers of fractional token holders economically viable; the same distribution executed on Ethereum mainnet would consume yields for low-value holdings. Third, Optimism’s EVM equivalence ensures full compatibility with the existing ERC-3643 contract suite and the Foundry toolchain used to deploy and test the contracts [13].

The on-chain contract suite consists of four primary components:

- 1) **RWAToken.sol**: An ERC-3643 compliant token contract that gates all transfers through a `canTransfer` pre-check against the `IdentityRegistry`.
- 2) **IdentityRegistry.sol**: An on-chain whitelist that maps investor wallet addresses to verified `ONCHAINID` claims [14].
- 3) **MultiSigEscrow.sol**: A 3-of-5 threshold multisignature contract that holds investor capital during the offering period.
- 4) **YieldDistributor.sol**: A Chainlink-triggered contract that enforces the revenue waterfall and executes USDC distributions.

IV. ECONOMIC MODEL AND REVENUE WATERFALL

A. Solar Asset Parameterization

The base unit of analysis is a single 450 W monocrystalline silicon photovoltaic panel. The capital cost of the complete hardware unit (panel, mounting, wiring, and meter) is R6,750 as of 2024 market rates in the South African context [1]. Table I summarizes the key input parameters for the economic model.

The base-case net yield of 20.7% per annum is computed after applying the 75/8/7/5/5 waterfall. The investor tranche of 75% of R2,014 yields R1,511 per annum against a capital cost of R6,750, producing a return on investment of 22.4% before tax. The protocol’s 5% platform fee and 5% expansion fund are retained on-chain, while the 8% operations and maintenance (O&M) fee and 7% insurance reserve are disbursed to the field operator (SunFix Logistics) and the reserve multisig wallet, respectively.

B. Sensitivity Analysis

The 20.7% base-case yield assumes constant irradiance of 5.5 sun-hours per day, zero panel degradation, and no inverter losses. Real-world conditions introduce variance from each

TABLE I
SOLAR ASSET ECONOMIC PARAMETERS (450 W PANEL, SOUTH AFRICA)

Parameter	Value	Unit / Note
Panel capacity	450	W
Capital cost	R6,750	ZAR (2024)
Avg. sun-hours	5.5	h day ⁻¹ (annual avg.)
Daily generation	2.475	kWh day ⁻¹
Annual generation	903.4	kWh year ⁻¹
EaaS rate	≤R2.23	ZAR/kWh (10% below Eskom)
Gross annual rev.	R2,014	ZAR (at R2.23/kWh)
Net yield (EaaS)	20.7%	of capital cost per annum

TABLE II
SENSITIVITY ANALYSIS: INVESTOR YIELD UNDER VARYING ASSUMPTIONS

Scenario	Sun-h	Deg.	Inv.	Yield
Conservative	4.0	0.7%	5%	13.8%
Low irradiance	4.5	0.5%	4%	16.2%
Base case	5.5	0%	0%	20.7%
Realistic adj.	5.5	0.5%	4%	18.7%
High irradiance	6.5	0.3%	3%	23.1%

Sun-h = daily sun-hours; Deg. = annual panel degradation (Year 1); Inv. = inverter loss.

of these assumptions. Standard monocrystalline silicon panels degrade at 0.5% to 0.7% per annum [15]; inverter efficiency losses are typically 3% to 5%. Table II presents the investor yield under five scenarios spanning conservative to optimistic assumptions.

Under the most conservative scenario (4.0 sun-hours, 0.7% degradation, 5% inverter loss), the investor yield drops to 13.8%, which remains positive but may fall below the risk premium required by retail investors in an emerging market context. The “realistic adjusted” scenario (5.5 sun-hours with standard degradation and inverter loss applied) produces an 18.7% yield, which we consider the most probable first-year return. The base-case 20.7% figure should therefore be understood as an upper bound on Year 1 performance assuming ideal conditions.

C. Seasonal Generation Curves

South Africa’s solar irradiance follows a pronounced seasonal pattern driven by its position in the Southern Hemisphere. Peak generation occurs during the austral summer months of November through February, when daily GHI averages 6.5 kWh m⁻² day⁻¹ in high-irradiance provinces such as the Northern Cape. The austral winter months of June and July represent the generation trough, with daily GHI dropping to approximately 4.5 kWh m⁻² day⁻¹ [3], [4].

Table III presents the estimated monthly generation and corresponding gross revenue for a 450 W panel across a representative annual cycle. The seasonal variation is a

TABLE III
ESTIMATED MONTHLY GENERATION AND GROSS REVENUE (450 W PANEL)

Month	Generation (kWh)	Gross Rev. (R)
January	101.5	226.4
February	91.7	204.5
March	83.8	186.9
April	68.9	153.6
May	57.1	127.3
June	49.5	110.4
July	52.3	116.6
August	64.3	143.4
September	75.6	168.6
October	88.4	197.1
November	98.7	220.1
December	72.6	161.9
Annual	903.4	R2,014

TABLE IV
REVENUE WATERFALL ALLOCATION (75/8/7/5/5)

Tranche	Recipient / Purpose	Share
1	Token holders (investor yield)	75%
2	Operations and maintenance (O&M)	8%
3	Insurance reserve fund	7%
4	Pool expansion (new arrays)	5%
5	Platform operations	5%

Note: all tranches denominated in USDC on Optimism Sepolia.

critical input for the yield engine: the billing cycle computation window is aligned to calendar months, and the `YieldDistributor` distributes the actual (not averaged) yield each month. Investors therefore receive higher distributions in summer months and lower distributions in winter months, consistent with the physical generation profile.

D. The 75/8/7/5/5 Revenue Waterfall

Gross revenue G computed per (2) is partitioned by the `YieldDistributor` contract into five tranches according to fixed percentage allocations defined in the contract’s immutable state. The waterfall is executed deterministically upon receipt of an authorized oracle report. Table IV defines each tranche and its economic rationale.

The 75% investor tranche is further modulated by a time-lock staking mechanism. Token holders who lock their ERC-3643 tokens in the `RWAVault` contract receive the full 75% allocation; liquid (unstaked) holders receive 60%, with the 15% differential returned to the depreciation reserve. This design encourages long-term capital commitment and reduces secondary market volatility.

The 7% insurance reserve is critical to the asset-backed nature of the protocol. In the event of panel failure, fire, or theft, insurance proceeds are routed through the SPV,

converted to USDC, and returned to token holders via the `YieldDistributor`'s liquidation path. The 5% expansion fund accumulates on-chain until a governance vote approves the acquisition of a new solar array, at which point it is routed to the `MultiSigEscrow` for a new capital formation cycle.

E. The 3-of-5 Multisignature Escrow Mechanics

During the initial offering period, investor USDC capital is held in the `MultiSigEscrow` contract. Capital release to the equipment seller requires the cryptographic confirmation of at least 3 out of 5 designated signers. The five signer roles are:

- 1) **Platform Operator:** Veridian protocol administrator.
- 2) **Solar Operator:** SunFix Logistics field representative.
- 3) **SPV Representative:** Director of the asset-holding SPV entity.
- 4) **Legal Representative:** Licensed conveyancer or commercial attorney confirming title transfer.
- 5) **Independent Verifier:** An auditor or regulatory observer confirming physical installation.

The 3-of-5 threshold is chosen to balance operational efficiency (a single party failure does not block the protocol) against security (any two parties acting collusively cannot release capital unilaterally). The contract is deployed with all five signer addresses registered at instantiation time; signer rotation requires a full 5-of-5 vote, making signer substitution an extraordinary governance action.

Capital is released from escrow in a single transaction only after the SPV Representative and Legal Representative have both signed, confirming off-chain legal title transfer. This two-role requirement ensures that the on-chain capital release is causally linked to the off-chain legal event, providing token holders with enforceable asset-backed collateral.

F. Oracle-Referenced Household Price Cap

To prevent investor governance from imposing extractive EaaS rates on lessee households, the protocol enforces a strict price cap constraint directly in the smart contract execution layer. The `PricingController` contract queries a Chainlink Price Feed that tracks the current regulated Eskom residential tariff (P_{grid}) and enforces:

$$P_{\text{proposed}} \leq 0.90 \times P_{\text{grid}} \quad (3)$$

Any DAO governance proposal that sets the EaaS rate above this ceiling is automatically reverted by the contract. This structural constraint guarantees that lessee households always pay at least 10% below the Eskom regulated tariff, preserving the economic incentive for households to adopt the EaaS model and protecting the protocol from the reputational and default risk of predatory pricing.

V. IMPLEMENTATION

A. Smart Contract Suite

The Veridian smart contract suite is implemented in Solidity 0.8.x and deployed using the Foundry toolchain on

Optimism Sepolia (Chain ID 11155420). The full contract source is available in the project repository at <https://github.com/sasssyboujee/veridian>. The four primary contracts are:

RWAToken.sol inherits from ERC-3643 and ERC20Votes. Every transfer and transferFrom call is intercepted by the `canTransfer` pre-check, which queries the `IdentityRegistry` via the `isVerified(address)` function. Transfers to non-whitelisted addresses are automatically reverted. The contract exposes `forcedTransfer` and `setAddressFrozen` functions accessible only to the designated Agent role, enabling compliance-mandated asset recovery without disrupting protocol state.

IdentityRegistry.sol maintains the triple registry structure required by ERC-3643: the `ClaimTopicsRegistry` (defining required claim types, e.g., KYC clearance, AML status, jurisdiction), the `TrustedIssuersRegistry` (enumerating authorized KYC providers), and the address-to-identity mapping. In the current testnet deployment, the `AutoKYC` contract acts as a mock claim issuer that directly injects an approved identity claim upon wallet submission. Production deployment requires replacement with a regulated KYC provider integration.

MultiSigEscrow.sol implements the 3-of-5 threshold logic described in Section IV-E. Investor USDC deposits are accumulated in the contract's token balance. A `confirmRelease` function increments a per-signer confirmation bitmap; when the confirmation count reaches 3, the `executeRelease` function becomes callable and transfers the full balance to the registered equipment seller address.

YieldDistributor.sol implements the waterfall logic. The `onReport` function, accessible only to the registered Chainlink `KeystoneForwarder` address, receives the ABI-encoded oracle payload containing the `net_yield_wei` scalar and a `token_snapshot` mapping of holder addresses to balances. The contract iterates over the snapshot, computing each holder's proportional allocation as:

$$Y_i = \frac{b_i}{B_{\text{total}}} \times 0.75 \times G_{\text{usdc}} \quad (4)$$

where b_i is the holder's token balance, B_{total} is the total supply, and G_{usdc} is the gross yield denominated in USDC-wei. The remaining 25% is partitioned into the four protocol tranches (8%, 7%, 5%, 5%) and transferred to their respective registered wallet addresses in the same transaction.

B. End-to-End Simulation Results

An end-to-end simulation was executed against the deployed testnet contracts using the `simulate_flow.py` script. The simulation exercised the complete protocol lifecycle: asset registration, billing cycle creation, TPM-signed telemetry ingestion, yield engine calculation, and oracle payload compilation.

Table V presents the key output values from the simulation run. The gross yield of R34,223.75 reflects a multi-cycle aggregation across a representative fleet segment. The net yield of R31,656.97, representing

TABLE V
END-TO-END SIMULATION OUTPUT (MULTI-CYCLE FLEET SEGMENT)

Metric	Value
Gross yield	R34,223.75
Champions fee (1%)	R342.24
Core fee (2%)	R684.48
Opportunity fee (4.5%)	R1,540.07
Total protocol fee	R2,566.78
Net yield	R31,656.97
Oracle payload (wei)	31,656,968,750,000,000,000,000

92.5% of gross yield after the 7.5% protocol fee split (Champions 1.0%, Core 2.0%, Opportunity 4.5%), was successfully encoded as a `net_yield_wei` value of 316569687500000000000000 (scaled to 18 decimal precision for EVM compatibility).

The simulation confirmed end-to-end pipeline integrity: the oracle payload was correctly structured for Chainlink Functions ingestion, the ECDSA telemetry signatures were verified against the registered public key, and the yield calculation matched the expected waterfall arithmetic to 18-decimal precision.

C. Token Issuance and ERC-3643 Compliance Flow

The issuance of fractional solar tokens follows a six-step compliance sequence. First, the investor submits their wallet address to the KYC portal. Second, the `AutoKYC` claim issuer (or, in production, a regulated provider) verifies the investor’s identity and issues a signed claim bound to their `ONCHAINID`. Third, the `IdentityRegistry` whitelists the investor’s address. Fourth, the investor deposits USDC into the `MultiSigEscrow`. Fifth, upon 3-of-5 confirmation and legal title transfer, the escrow releases capital and the `RWAFactory` deploys a new `RWAToken` instance and mints fractional tokens proportional to each investor’s contribution. Sixth, the `RWAFactory` also deploys a dedicated `RWAGovernor` DAO contract for the asset, granting token holders voting power proportional to their fractional ownership.

D. Gas Cost and Throughput Evaluation

The `YieldDistributor` contract uses a push-based distribution pattern that iterates over all token holders in a single transaction. This design has $O(n)$ gas complexity, where n is the number of holders. On Optimism, each USDC transfer within the loop consumes approximately 26,000 gas (21,000 base + 5,000 for ERC-20 storage writes). The total gas cost per distribution cycle is therefore:

$$C_{\text{push}}(n) = C_{\text{base}} + n \times C_{\text{transfer}} \quad (5)$$

where $C_{\text{base}} \approx 50,000$ gas (function dispatch, oracle verification, waterfall split) and $C_{\text{transfer}} \approx 26,000$ gas per holder.

TABLE VI
PUSH DISTRIBUTION GAS COST ON OPTIMISM (L2)

Holders	Gas	Est. Cost	Break-even
10	310,000	\$0.001	R0.02/holder
100	2,650,000	\$0.008	R0.17/holder
500	13,050,000	\$0.039	R0.85/holder
1,000	26,050,000	\$0.078	R1.70/holder
5,000	130,050,000	\$0.390	R8.50/holder

Break-even = min. monthly yield per holder for gas cost not to exceed 1% of distribution.

Table VI presents the estimated gas consumption and USD cost at an Optimism L2 gas price of 0.001 Gwei.

At 1,000 holders, the push distribution remains economically viable on Optimism (\$0.078 per cycle). However, two scalability constraints emerge. First, the Optimism block gas limit (30M gas) imposes a hard ceiling of approximately 1,150 holders per single distribution transaction. Second, for holders with very small fractional positions, the per-holder gas cost may exceed the yield distributed.

A Merkle-proof pull distribution pattern provides an $O(1)$ alternative. In this model, the contract administrator publishes a Merkle root of the yield allocation tree, and each holder independently submits a proof to claim their yield. The gas cost per claim is approximately 50,000 gas regardless of the total holder count, with the trade-off that unclaimed yields accumulate until the holder initiates a claim transaction. The pull pattern is recommended for deployments exceeding 1,000 holders and is planned for future protocol versions.

VI. EVALUATION

A. Security Analysis

The protocol defines five quantitative circuit-breaker thresholds. Breach of any threshold triggers automated protective action:

- 1) **Telemetry Variance** ($\leq 5\%$): Discrepancy between hardware-signed generation data and physical audit findings triggers operator stake slashing.
- 2) **Secondary Market Liquidity Ratio** ($\geq 15\%$ of market cap within 30 days): Failure indicates inadequate secondary market depth; automated circuit breaker halts new token issuance.
- 3) **Governance Quorum Floor** ($\geq 10\%$ of circulating supply): Failure on critical votes triggers automatic fallback to the Maintenance Council, preventing operational paralysis.
- 4) **Price Cap Compliance**: Any proposed rate exceeding $0.90 \times P_{\text{grid}}$ (equation 3) is automatically reverted by the `PricingController`.
- 5) **Yield-to-CapEx Accumulation**: The expansion fund requiring more than 24 months to accumulate a 20% down payment on a new mid-tier array signals stalled network growth and triggers a governance review.

TABLE VII
COMPARISON WITH RELATED ENERGY BLOCKCHAIN PROTOCOLS

Feature	Veridian	Power Ledger	Ondo Finance
Hardware attestation.	TPM 2.0 ECDSA	None	None
Investor KYC	ERC-3643 on-chain	Off-chain	Centralized
Settlement	USDC (L2)	POWR token	USDC (main-net)
Asset type	Solar (EaaS)	PV	Grid energy
Household disc.	$\geq 10\%$ vs. grid	Market rate	N/A
Multisig escrow	3-of-5	None	Custodial
Yield profile	Seasonal	Real-time	Fixed

The hardware attestation layer provides a strong security guarantee for the telemetry data. Because the TPM’s private key is hardware-resident and cannot be exported, an adversary seeking to inflate reported generation must either compromise the physical meter hardware or conduct a man-in-the-middle attack between the meter and the backend API. The ECDSA verification in the backend (equation 1) and the 5% variance threshold provide complementary defenses against these attack vectors.

B. Comparison with Existing DePIN Energy Protocols

Table VII compares Veridian with two representative existing protocols in the energy DePIN and RWA tokenization space.

Veridian is distinguished from Power Ledger by the hardware attestation layer and the ERC-3643 compliance mechanism. Power Ledger requires functioning grid interconnection and does not provide regulated fractional investment instruments. Veridian is distinguished from Ondo Finance by the physical asset class (solar PV infrastructure versus public securities), the hardware attestation-grounded telemetry pipeline, and the explicit household protection constraint in equation (3).

C. Token Standard Suitability

Table VIII compares the three principal Ethereum token standards considered for the fractional solar tokens.

ERC-3643 [10] was selected because it enforces compliance at the smart contract execution layer, making it impossible for a non-whitelisted wallet to receive solar asset tokens regardless of the calling mechanism. This property is essential for South African FSCA compliance requirements for retail investment products.

D. Trust Model and Limitations

The Veridian prototype operates under a *two-tier trust model* that distinguishes between design-level security guarantees and the current evaluation-level trust assumptions. This distinction is critical for an accurate assessment of the protocol’s maturity.

TABLE VIII
TOKEN STANDARD COMPARISON FOR REGULATED SOLAR ASSET TOKENS

Standard	Compliance	Suitability
ERC-20	None (permissionless)	Unsuitable: cannot enforce KYC/AML at transfer layer
ERC-1400	Static role-based	Partial: high gas cost, complex partition management
ERC-3643	Dynamic registry pre-check on every transfer	Optimal: enforces KYC/AML, supports forced transfer, aligns with securities regulation

Tier 1: Hardware Attestation (Design-level guarantee; currently simulated). The protocol design specifies that telemetry integrity is anchored to a TPM 2.0 hardware root of trust whose private key is non-exportable. Under physical deployment, an adversary seeking to forge generation data must compromise the physical meter hardware itself, reducing the threat surface to physical tampering rather than software exploitation. However, in the current prototype, the TPM is simulated using software-generated SECP256R1 keys. The signing key is therefore extractable, and the hardware attestation guarantee is *unverifiable* at the evaluation level. The development bypass in `tpm_verify.py` further weakens this tier by permitting unverified payloads. The threat surface reduction that physical TPM deployment would achieve is bounded by the TPM 2.0 specification’s key non-exportability guarantee [12]; the current simulation validates the protocol flow and payload format, but not the tamper-resistance property.

Tier 2: Oracle Delivery (Decentralized consensus guarantee; currently centralized). The protocol design specifies yield distribution triggered by a Chainlink Functions decentralized oracle network, where multiple independent nodes reach consensus on the yield payload before submission to the `YieldDistributor`. In the current prototype, this delivery is simulated by a single FastAPI backend making a direct contract call. This means the current system is architecturally equivalent to a centralized platform distributing funds via an API call. The “deterministic settlement” claim therefore applies to the *contract-level execution logic* (which enforces the waterfall regardless of the caller) but not to the *trigger mechanism*, which currently lacks decentralized consensus.

Both tiers must be independently evaluated when assessing the protocol’s security posture. The current prototype operates at the weakest level of both tiers: simulated hardware attestation and centralized oracle delivery.

In addition to the trust model limitations, two further components require production-grade replacement:

- 1) **AutoKYC (Mocked):** The `AutoKYC` claim issuer bypasses identity verification. Production deployment requires integration with a regulated KYC provider (e.g., Onfido, Jumio) connected to the FSCA’s AML/CFT

reporting framework.

- 2) **Midnight Network zk-KYC (Proposed):** The integration of Midnight Network zero-knowledge proofs for selective identity disclosure is architecturally specified but not implemented. This component would enable investors to prove KYC clearance without exposing personal data to the public ledger.

VII. DISCUSSION

A. Structural Disintermediation of Grid Monopoly

The Veridian protocol achieves grid-monopoly disintermediation through a combination of physical infrastructure bypass and financial democratization, rather than through regulatory confrontation. By providing households with solar energy via the EaaS model at a rate contractually bounded below the Eskom regulated tariff (equation 3), the protocol reduces the household's dependence on grid supply without requiring grid disconnection. As panel fleets scale, the aggregate distributed generation capacity represents a meaningful reduction in peak demand that Eskom would otherwise need to supply through thermal generation.

From the investor perspective, the protocol provides access to an asset class (physical solar infrastructure) that has historically been accessible only to utilities, pension funds, and infrastructure private equity. Fractional tokenization at the single-panel level (R6,750 capital cost per unit) reduces the minimum investment to a retail scale. The ERC-3643 compliance layer ensures that this democratization does not come at the cost of regulatory compliance or investor protection.

B. Scalability Pathway

The current simulation operates at a single-panel granularity. Fleet-scale deployment would aggregate multiple panels into a single SPV and issue tokens representing a proportional share of the fleet's combined generation. The yield engine's `billing_cycles` schema supports multi-asset aggregation natively; the primary technical barrier to scale is the cost of on-chain token snapshot iteration in the `YieldDistributor`. At large holder counts, the gas cost of the push-distribution loop could become prohibitive even on Optimism. This motivates a future migration to a Merkle-proof-based pull distribution model, where holders claim their yields by submitting a proof rather than receiving an automatic airdrop.

C. Regulatory Pathway for South Africa

The SPV and token structure described in this paper constitute a *design-level legal assumption* that has not been validated by a formal legal opinion. The following analysis identifies the relevant South African legislation and the specific licensing requirements that would apply.

Under the Financial Markets Act 19 of 2012, instruments that offer profit participation derived from a third party's operational efforts are classified as securities. The Veridian EaaS tokens, which provide yield derived from SunFix Logistics' operation of the solar panels, would likely satisfy this classification. Additionally, the Collective Investment Schemes

Control Act 45 of 2002 (CIS Act) may apply if the FSCA determines that pooled investor capital managed by an SPV constitutes a collective investment scheme, which would require a Section 5 license from the FSCA.

The SPV itself would be incorporated under the Companies Act 71 of 2008 as a private company (Pty. Ltd.) with the singular purpose of holding legal title to the solar panel fleet. The ERC-3643 standard's mandatory KYC/AML enforcement and the `IdentityRegistry` provide the on-chain compliance infrastructure that the FSCA would require for a retail investment product.

However, three off-chain legal prerequisites remain unresolved: (i) formal FSCA registration as a CIS, requiring a licensed management company and approved trust deed; (ii) a legal opinion confirming that the ERC-3643 token transfer constitutes a valid transfer of beneficial interest under South African law; and (iii) compliance with the Financial Intelligence Centre Act (FICA) for customer due diligence. These off-chain legal structures are outside the scope of the current technical prototype and represent the primary pathway dependency for production deployment. No formal legal opinion has been obtained, and the regulatory analysis presented here should not be construed as legal advice.

VIII. CONCLUSION

This paper has presented Veridian, a DePIN protocol for hardware-attested fractional financing of solar infrastructure in South Africa. The protocol makes two primary contributions to the DePIN and decentralized energy systems literature.

At the *design level*, the protocol specifies an end-to-end trust chain from TPM 2.0 hardware-signed telemetry through an ERC-3643 compliance-gated token layer to deterministic USDC yield settlement on Optimism, governed by the 75/8/7/5/5 revenue waterfall and secured by a 3-of-5 multi-signature escrow. The oracle-referenced price cap mechanism ($P_{\text{proposed}} \leq 0.90 \times P_{\text{grid}}$) represents a novel governance constraint that structurally prevents investor-captured DAOs from extracting surplus from energy consumers.

At the *evaluation level*, the current prototype operates under a two-tier trust model (Section VI-D): hardware attestation is simulated with software keys, and oracle delivery is centralized. The end-to-end simulation validates the economic model and protocol flow: a 450 W panel produces a base-case investor yield of 20.7% per annum (18.7% under realistic degradation and inverter loss assumptions per Table II), with seasonal generation bounded between 49.5 kWh in June and 101.5 kWh in January. The gas cost analysis (Section V-D) confirms that push-based yield distribution is viable for up to approximately 1,150 holders on Optimism before reaching the block gas limit, motivating a Merkle-proof pull pattern for larger deployments.

The path from prototype to production requires resolving four dependencies: (i) physical TPM 2.0 provisioning in field metering hardware, (ii) regulated KYC provider integration replacing the mocked AutoKYC, (iii) Chainlink Functions mainnet subscription for decentralized oracle delivery, and

(iv) formal FSCA registration under the CIS Act (Section 45 of 2002) and Financial Markets Act (19 of 2012). Each dependency is independently addressable and does not require architectural changes to the protocol design.

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REFERENCES

- [1] National Energy Regulator of South Africa, “State of the Electricity Distribution Industry Report 2022/23,” NERSA, Pretoria, South Africa, Tech. Rep., 2023.
- [2] World Bank, “Tracking SDG 7: The Energy Progress Report 2023,” World Bank Group, Tech. Rep., 2023.
- [3] South African Weather Service, “South Africa Solar Resource Atlas: Solar Irradiation Data 2023,” South African Weather Service, Pretoria, South Africa, Tech. Rep., 2023.
- [4] International Renewable Energy Agency, “South Africa: Renewable Energy Country Profile,” IRENA, Tech. Rep., 2022.
- [5] Eskom Holdings SOC Ltd, “Tariff and Charges Booklet 2024/25,” Eskom, Johannesburg, South Africa, Tech. Rep., 2024.
- [6] C. Pop, I. Ciurea, M. Benta, and M. Moldovan, “Peer-to-peer energy trading in microgrids based on ethereum blockchain,” in *Proc. IEEE International Conference on Smart Energy Systems and Technologies (SEST)*, 2018, pp. 1–6.
- [7] M. Andoni, V. Robu, D. Flynn, S. Abram, D. Geach, D. Jenkins, P. McCallum, and A. Peacock, “Blockchain technology in the energy sector: A systematic review of challenges and opportunities,” *Renewable and Sustainable Energy Reviews*, vol. 100, pp. 143–174, 2019.
- [8] L. Zhang, W. Liu, and R. Chen, “Decentralized physical infrastructure networks: Architecture, incentive design, and security analysis,” in *Proc. IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 2022, pp. 1–9.
- [9] M. Massaro, E. Jahani, and R. Srivastava, “Real-world asset tokenization: Regulatory challenges and technical architectures,” *Journal of Financial Regulation*, vol. 9, no. 1, pp. 55–89, 2023.
- [10] Tokeny Solutions, “EIP-3643: T-REX – Token for Regulated EXchanges,” Ethereum Foundation, Tech. Rep., 2021, ethereum Improvement Proposal 3643. [Online]. Available: <https://eips.ethereum.org/EIPS/eip-3643>.
- [11] A. Boudguiga, N. Bouzerna, L. Granboulan, A. Olivereau, F. Raynal, A. Roger, and R. Sirdey, “Towards better availability and accountability for IoT updates by means of a blockchain,” in *Proc. IEEE European Symp. Security and Privacy Workshops (EuroS&PW)*, 2017, pp. 50–58.
- [12] Trusted Computing Group, “Trusted Platform Module Library Specification, Family “2.0”, Level 00, Revision 01.83,” Trusted Computing Group, Tech. Rep., 2024, [Online]. Available: <https://trustedcomputinggroup.org/resource/tpm-library-specification/>.
- [13] OP Labs, “Optimism Documentation: Bedrock Architecture,” OP Labs, Tech. Rep., 2023, [Online]. Available: <https://docs.optimism.io>.
- [14] Tokeny Solutions, “ONCHAINID: A Decentralized Identity Protocol for Regulated Tokens,” Tokeny Solutions, Tech. Rep., 2022, [Online]. Available: <https://docs.onchainid.com>.
- [15] E. Vartiainen, G. Masson, C. Breyer, D. Moser, and E. R. Medina, “Impact of weighted average cost of capital, capital expenditure, and other parameters on future utility-scale PV levelised cost of electricity,” *Progress in Photovoltaics: Research and Applications*, vol. 28, no. 6, pp. 439–453, 2020.